



Daeun Kim

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Research Interests

keywords: Software Supply Chain Security, Open-Source Software Security, Static Analysis, Vulnerability Detection, Software Composition Analysis

Education

Kyung Hee University	Gyeonggi-do, South Korea
B.S. in Software Convergence, College of Software	Mar. 2024 - Feb. 2027 (Expected)
• GPA: 4.23/4.5 (Major of: 4.23/4.5)	

Experience

Undergraduate Intern	Gyeonggi-do, South Korea
SIFT Lab, Kyung Hee University	Jan. 2026 - present
• Conducted research on software security, including a software supply chain risk measurement study and static analysis for preserving referential integrity in synthetic data.	

Publications

Improving Referential Integrity of Multi-Table Synthetic Data via Static Analysis of Eloquent ORM Relationships	
Da-Eun Kim, Korea Computer Congress (KCC), 2026.	2026
• Proposed a static analysis method to generate referentially-consistent synthetic databases for realistic security testing environments, addressing foreign-key integrity loss in existing synthetic data tools.	
• Implemented a Python-based parser extracting PK-FK relationships from Laravel Eloquent ORM code (belongsTo, hasMany, hasOne, belongsToMany) and converted them into SDV multi-table metadata .	
• Achieved 99.73% precision, 98.95% recall , and 99.34% F1-score in relationship extraction across three open-source Laravel projects, and validated that the extracted metadata eliminates foreign-key violations in generated synthetic test data.	

Projects

Update-Delta Analysis of Trust-Sensitive Behavior in VS Code Extensions, Kyung Hee Univ.	May. 2026 - Present
• Conducted a measurement study on the Open VSX extension ecosystem , collecting 1,000 update pairs (previous vs. latest versions) to analyze security risk changes introduced through updates.	
• Built a Python-based static analysis pipeline extracting trust-sensitive behaviors (child_process, file-system/network access, eval, postinstall scripts, dependency changes) and defined risk_score and risk_delta metrics to quantify update-induced risk.	
• Applied statistical tests (Kruskal-Wallis, Mann-Whitney U, Chi-square) across download count, category, and update-gap length, finding that longer update gaps were significantly associated with larger risk-score changes.	
• Currently extending the analysis to full version-history sequences (beyond single update pairs) to detect suspicious update patterns indicative of malicious behavior in the software supply chain.	

AI-Powered Asset Management Service “GrowPlan”, 9roomthon Univ

Aug. 2025 - Sep. 2025

Front-End Developer

Tech Stack : React, Vite, TailwindCSS

Link : https://github.com/dannnnne/2025_SEASONTHON_TEAM_35_FE

- Developed the **front end of an AI-powered asset management service** that helps young adults analyze their financial structure and plan asset allocation based on personal financial goals.
- Built core **user workflows** including Kakao-based authentication, asset management, goal setting, progress tracking, and user profile management.
- Implemented **data visualizations for users' current asset composition and AI-recommended allocation** across investment, savings, and emergency funds.

AI-Powered UX/UI Freelance Matching Platform, Kyung Hee University

May. 2025 - Oct. 2025

Front-End Developer

Tech Stack : React, TypeScript, TailwindCSS

Link : <https://morak-frontend.vercel.app/>

- Led the **end-to-end front-end development** of an AI-powered UX/UI freelance matching platform designed to reduce requirement ambiguity and communication gaps between clients and designers.
- Implemented the front-end UI for an **AI-assisted requirement refinement workflow** with dynamically generated follow-up questions and hierarchical question-answer flows based on client responses.
- Built core **service workflows** including project creation and management, designer matching, chat, portfolio management, and intermediate and final deliverable handling, integrated with backend REST APIs to manage complex project states and AI-generated data.

Activities

9oormthon Univ 4th Cohort, Kyung Hee Univ.

May. 2025 - Nov. 2025

- Strengthened technical skills through in-school study sessions and developed a web project.
- Developed another web project through a 2-day overnight hackathon, from ideation to demo day.

Honors

Semester High Honors, Kyung Hee Univ.

2025-2

- Awarded to students with high achievements throughout the semester

Skills

Programming : Python, C++, JavaScript, TypeScript, PHP

Frameworks/Libraries: React, TailwindCSS

Languages

Korean: Native

English: Intermediate High (TOEIC Speaking 150)